Vaani Market

A voice-first assistant that explains the Indian stock market in Telugu, Hindi and English. It explains. It never advises.

The problem

Following a company means reading quarterly results, working out why a price moved, and learning terms like P/E or circuit limit. For a student or a working person that research takes more time than they have, and most of it is written in English. A person can ask "why did GAIL fall today?" out loud in their own language without knowing any of the vocabulary.

The solution

Speak a question in Telugu, Hindi or English and Vaani Market answers aloud in the same language, in a phone-friendly web app.

- **Beginners** ask what a stock, NSE, BSE or a candlestick is. They get answers from a human-reviewable glossary, with an illustrative diagram.
- **Light traders** ask what happened today for HDFC, GAIL, IOC, GoldBEES, Nifty or Sensex. They get answers from live (delayed) data and timestamped headlines, with a 30-day chart.

How it works

Gnani **Prisma v3.0** turns speech into text and detects the language. A planner (one constrained LLM call that returns JSON only) picks the intent. A ticker resolver maps the spoken name to a symbol, and asks "which HDFC?" when a name is ambiguous. Plain code then fetches the price, trend and headlines, and every figure carries its source and timestamp. Where needed, a tiered retrieval step (keyword glossary, then hybrid BM25 + dense search) adds context. A short three-sentence explanation is written from the fetched facts only, a numeric gate checks it, and Gnani **Timbre v3.0** speaks it sentence by sentence.

Why it can be trusted

- **It never advises.** A deterministic classifier runs on the question before any LLM call, and again on the drafted answer. The refusal is a fixed, pre-translated message, never generated. No order-placing tool exists anywhere in the code.
- **Numbers come from code, not the model.** Every number in an answer is checked against the fetched data. A mismatch triggers one regeneration, then a safe template.
- **It is honest about time and cause.** It says "latest available as of ..." when the market is closed or the data is delayed, and says plainly when the reason for a move is unknown.
- **Retrieved text is treated as data.** A test headline telling it to recommend BUY did not change its behaviour.

Evidence

A 40-case adversarial suite (direct, indirect and role-play advice, ambiguous tickers, code-mixed speech, out-of-scope and missing data) passes 40 of 40 after the fixes the harness itself found. 63 offline tests cover the safety-critical logic. All published figures trace to `eval/results/metrics.csv`, and anything unmeasured is labelled as a target.

Gnani models and reasoning layer

Speech to text: **Prisma v3.0**. Text to speech: **Timbre v3.0**. Gnani offers no language-model endpoint, so the reasoning layer is a lightweight Gemini Flash-Lite model behind a swappable adapter.

Honest limits

Market data is delayed and unofficial. Telugu and Hindi text are drafts pending native-speaker review (`docs/TELUGU_REVIEW.md`). Advanced retrieval (Tier 3, graph) is designed but not built, and retrieval quality in Telugu and Hindi is unmeasured. This is an engineering demonstration, not legal or investment advice.

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